

Sub: _____

Part 3: Gradient Descent

Date: _____

Manual Calculation

Part 1 (Iteration 1)

Given information:

$$y = m_1 x_1 +$$

$$\text{initial } m = [-1, 2]$$

$$b = [1, 1]$$

$$x = [(1, 3), (4, 10)]$$

$$y = (5, 6)$$

$$n = 2$$

Learning rate: 0.01 (Supposed)
 $\alpha =$

$$x = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}, m = \begin{bmatrix} -1 \\ 2 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

Step 1: ~~predict~~ ^{Calculate} y-predict

$$y\text{-predict} = x m + b = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} (1)(-1) + (3)(2) \\ (4)(-1) + (10)(2) \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 16 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}$$

$$\text{error vector } e = y\text{-predict} - y = \begin{bmatrix} 6 - 5 \\ 17 - 6 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

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$$MSE = \frac{\sum (y_{\text{predict}} - y)^2}{n}$$

$$= \frac{1^2 + 11^2}{2} = \frac{122}{2} = 61$$

Step 2: Finding the gradient of MSE.

$$U(m, b) = \frac{1}{n} (X_m + b - y)^T (X_m + b - y)$$

$$\text{let } u = X_m + b - y$$

$$E = \frac{u^2}{n}$$

$$\frac{dE}{du} = \frac{2u}{n}$$

$$\frac{du}{dm} = X^T e = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ 11 \end{bmatrix} =$$

$$= \begin{bmatrix} (1)(1) + (4)(11) \\ (3)(1) + (10)(11) \end{bmatrix} = \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

Step 3: Update m and b (first iteration)

$$m^{(1)} = m - \alpha \frac{du}{dm} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} - 0.01 \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

$$= \begin{bmatrix} -1 - 0.45 \\ 2 - 1.13 \end{bmatrix} = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$$

$$b^{(1)} = b - \alpha \frac{du}{db} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 0.01 \begin{bmatrix} 1 \\ 11 \end{bmatrix} = \begin{bmatrix} 1 - 0.01 \\ 1 - 0.11 \end{bmatrix}$$

$$= \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}$$

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So After the 1st iteration,

new data:

$$m = [-1.45, 0.87]$$

$$b = [0.99, 0.89]$$